



Osteoarthritis Biomarkers Consortium FNIH Project: Measurement of Subchondral Bone Integrity from Knee Radiographs by Duke Image Analysis Laboratory

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1. Overview

1.1 SAS dataset

Name: **kXR_FNIH_BTI_DukeXX** (XX identifies the time point)

Display label: **kXR FNIH Bone Trabecular Integrity (Duke)**

1.2 Contents of dataset

These datasets contains measurements of subchondral bone trabecular integrity (BTI) for 600 knees from the OA Biomarkers Consortium FNIH Project. The analysis of these images was performed at Duke University under the supervision of Virginia Kraus and H Cecil Charles.

IMPORTANT NOTE: Since the initial release of these datasets, the BTI parameters were recalculated from the same raw fractal signature analysis data using the method described in this publication:

- *Deveza, L.A., et al (2016) "The association between biochemical markers of bone turnover and bone changes on imaging – Data from the Osteoarthritis Initiative". Arthritis Care & Research (in press) <http://dx.doi.org/10.1002/acr.23121>*

Each dataset contains the reading data for a single visit, or time point. The corresponding longitudinal data for other time points are in separate datasets. For example, the data from this vendor's reading of baseline x-rays are in a dataset ending in "00" (i.e., kMRI_FNIH_BTI_Duke00). The corresponding data from this vendor for the 12-month visit MRI readings are in a dataset with the same name but ending in "01" (i.e., kXR_FNIH_BTI_Duke01). The corresponding data from this vendor for the 24-month visit readings are in a dataset with the same name but ending in "03" (i.e., kMRI_FNIH_BTI_Duke03).

See the document "VisitPrefixDefinitions.pdf" document for a guide to visit numbering.

Data from baseline, 12-month and 24-month visits are available. To compare values of a variable across time points from a given project by this vendor, or to calculate change scores, users will need to merge the datasets for the various time points.



1.3 Condition

- Known data errors: none at present.
- Dataset strengths/weaknesses:
 - o Data are expected for all participants included in a project sample. If expected data do not exist for a knee, SAS special missing values are assigned to denote why the data were not obtained.
 - o The dataset contains one row of data (record) for a given knee which needs to be taken into account when merging it with other datasets. Please see the document "*Overview and Description of Central Image Assessments*" for more information on merging.
 - o The variable ID can be used for merging with the OA Biomarkers Consortium FNIH Project Clinical dataset (Clinical_FNIH) and OAI publically released datasets that contain one record per participant.
 - o To merge with other OA Biomarkers Consortium FNIH Project datasets of image assessments, and with OAI publically released datasets of image assessments (which may contain more than one record per participant), the variables ID and SIDE must be used to correctly match up data for the correct knee (left or right).

1.4 Variables and reading methods

A complete listing of the variables in these datasets can be found in the documentation provided with the dataset, including SAS variable names, descriptive variable labels and attributes.

Variables assessed for all projects by this vendor include:

- 3 extracted variables from fractal signature plots (fractal dimension versus radius) generated by the vertical filter of the KneeAnalyzer software from Optasia Medical from plain knee radiographs
- 3 extracted variables from fractal signature plots (fractal dimension versus radius) generated by the horizontal filter of the KneeAnalyzer software from Optasia Medical from plain knee radiographs
- The vertical filter data reports on the horizontal (tensile) trabecular integrity
- The horizontal filter data reports on the vertical (compressive) trabecular integrity

These publications give more details about the methods used for making these quadratic fits:

- Devez, L.A., et al (2016) "The association between biochemical markers of bone turnover and bone changes on imaging – Data from the Osteoarthritis Initiative". *Arthritis Care & Research* (in press) <http://dx.doi.org/10.1002/acr.23121>
- Kraus, V. B., et al. (2009). "Trabecular morphometry by fractal signature analysis is a novel marker of osteoarthritis progression." *Arthritis Rheum* 60(12): 3711-3722. PMID: 19950282 PMCID: PMC3711179 <http://dx.doi.org/10.1002/art.25012>
- Kraus, V. B., et al. (2013). "Subchondral bone trabecular integrity predicts and changes concurrently with radiographic and magnetic resonance imaging-determined knee osteoarthritis progression." *Arthritis Rheum* 65(7): 1812-1821. PMID: 23576116 PMC4152231 <http://dx.doi.org/10.1002/art.37970>



The measurements of fractal signatures from which the quadratic fits were calculated were done using Optasia Medical's Knee Analyser software (<http://www.optasiamedical.com/products/knee-tool/>) and the method which that software uses for calculating fractal signatures has been published:

- Lynch, J. A., et al. (1991). "Analysis of Texture in Macroradiographs of Osteoarthritic Knees Using the Fractal Signature." *Physics in medicine and biology* 36(6): 709-722. PMID: 187120 <http://dx.doi.org/10.1088/0031-9155/36/6/001>
- Lynch, J. A., et al. (1991). "A Robust and Accurate Method for Calculating the Fractal Signature of Texture in Macroradiographs of Osteoarthritic Knees." *Medical Informatics* 16(2): 241-251. PMID: 1921566



2. Methods specific to this Project

2.1 Image type

Bilateral PA Fixed Flexion weight-bearing knee radiographs were used, and details of the acquisition protocol can be found at the “*Radiography (X-ray) Manual*.”

2.2 Time points

Baseline, 12-month, and 24-month visits.

2.3 Measurement methods

Prior to transferring images to Duke Image Analysis Laboratory, the x-ray images were blinded to both the OAI Release ID and time point of acquisition, and also masked the bilateral x-ray so that the contralateral knee was not visible to the readers.

2.4 Variables

Each dataset contains variables for identifying the ID and which knee was analyzed, along with a unique 16-digit number identifying the OAI x-ray analyzed. The BTI measurements then comprise a total of 6 variables generated from analysis of the medial tibial subchondral bone. Extracted variables from the fractal signature plots are as follows:

- (i) a vertical filter intercept value
- (ii) a vertical filter linear slope value
- (iii) a vertical filter quadratic slope value
- (iv) a horizontal filter intercept value
- (v) a horizontal filter linear slope value
- (vi) a horizontal filter quadratic slope value

These variables have been extracted in such a way as to be orthogonal parameters, i.e., independent of one another. They can therefore be evaluated individually and in combination for their predictive capability. This is accomplished by deriving the linear and quadratic terms from the nadir of the fractal dimension curves. The overall ability to use BTI as a predictive measure is not impacted, only improved in that the parameters are not a priori collinear. The parameters reported for the OA Biomarkers Consortium FNIH Project represent parameters that are orthogonal. The absolute values of the parameters are therefore not directly comparable to prior published work but ability to be predictive of outcomes should be equivalent

A complete listing of the variables in these datasets can be found in the documentation provided with the dataset, including SAS variable names, descriptive variable labels and attributes.

2.5 Sample

Details of the selection process for the 600 knees analyzed for this project are given in the documentation for the OA Biomarkers Consortium FNIH Project Clinical dataset (Clicnial_FNIH_Descrip.pdf).

3. References

1. Deveza, L.A., et al (2016) “The association between biochemical markers of bone turnover and bone changes on imaging – Data from the Osteoarthritis Initiative”. *Arthritis Care & Research (in press)*
<http://dx.doi.org/10.1002/acr.23121>
2. Kraus, V. B., et al. (2009). “Trabecular morphometry by fractal signature analysis is a novel marker of osteoarthritis progression.” *Arthritis Rheum* 60(12): 3711-3722. PMID: 19950282 PMCID: PMC3711179
<http://dx.doi.org/10.1002/art.25012>



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3. Kraus, V. B., et al. (2013). "Subchondral bone trabecular integrity predicts and changes concurrently with radiographic and magnetic resonance imaging-determined knee osteoarthritis progression." *Arthritis Rheum* 65(7): 1812-1821. PMID: 23576116 PMC4152231 <http://dx.doi.org/10.1002/art.37970>
 4. Lynch, J. A., et al. (1991). "Analysis of Texture in Macroradiographs of Osteoarthritic Knees Using the Fractal Signature." *Physics in medicine and biology* 36(6): 709-722. PMID: 187120
<http://dx.doi.org/10.1088/0031-9155/36/6/001>
 5. Lynch, J. A., et al. (1991). "A Robust and Accurate Method for Calculating the Fractal Signature of Texture in Macroradiographs of Osteoarthritic Knees." *Medical Informatics* 16(2): 241-251. PMID: 1921566